Ashish Gajjela

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EDUCATION

Northeastern University, Khoury College of Computer Science — Boston, MA
Master of Science in Artificial Intelligence, ***GPA: 4.0 / 4.0*** — **Expected:** May 2027

Jawaharlal Nehru Technological University — Hyderabad, India
Bachelor of Technology in Computer Science and Engineering (AI & ML), ***GPA: 8.14 / 10.0*** — December 2020 – May 2024

Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Machine Learning Operations (MLOps), Reinforcement Learning, Natural Language Processing, Computer Vision, Programming Design Paradigm, Algorithms

EXPERIENCE

Eizen (Contractor for Rakuten SixthSense Cognitive AI) — Hyderabad, India
Machine Learning Engineer Intern — January 2024 – November 2024

- Designed and trained a **time-series fusion transformer** enhanced with **Neural Architecture Search (NAS)**, improving anomaly detection accuracy by **18%** on large-scale multivariate temporal datasets.
- Developed a **Graph Attention Network (GAT)** to model cross-entity dependencies, reducing **root cause analysis time by 60%**.
- Assembled containerized inference services using **FastAPI** and **Kubernetes** on **GCP**, enabling automated execution of **100K+ anomaly evaluations/day** with robust logging, monitoring, and CI/CD pipelines.
- Architected a **multi-agent conversational system** using LangGraph, orchestrating **tool-using agents** to retrieve and reason over **real-time observability data** (metrics, anomalies), enabling SREs to ask natural language questions and receive actionable answers.

EpsilonPi (Student Club) — Hyderabad, India
Generative AI Developer — August 2022 – June 2023

- **Prototyped** and benchmarked **LLM-driven conversational AI systems** using **FLAN-T5** and **HuggingFace Transformers**, improving assistant usability by **20%**.
- **Systematized** reusable pipelines for data preprocessing, fine-tuning, and deployment using **Python** and **NLTK**, reducing end-to-end prototyping time by **30%** across student research projects.
- Investigated **dialogue generation behavior in LLMs**, uncovering hallucination patterns and fine-tuning constraints, and distilled findings into actionable guidance for future experimentation.

TECHNICAL LEADERSHIP

Co-Founder & CTO, Twinly | website — March 2025 – July 2025

- Co-founded **Twinly**, an AI-powered productivity platform automating task management across Gmail, Notion, and Slack using secure API integrations, reducing user context-switching by **35%**.
- Architected distributed microservices using **FastAPI**, **PostgreSQL**, **AWS EC2/Lambda/S3**, and asynchronous event-driven workflows, achieving **98% uptime** with low-latency synchronization across cloud environments.
- Configured **OAuth2** and **JWT**-based authentication with CI/CD-enforced security checks, audit logging, and telemetry dashboards (Prometheus/Grafana) to ensure fault-tolerant, compliant system behavior.
- Orchestrated and fine-tuned **Phi-3 agentic AI workers** with telemetry-driven feedback loops, improving task relevance, memory recall, and multi-step action reliability by **30%**.
- Led a cross-functional team of four engineers and designers through agile sprints, delivering the MVP **three weeks early** and documenting end-to-end system architecture for scalable product iteration.

PROJECTS

ADK Driven Multi-Agent Conversational AI System | Python, ADK, Qwen

- Architected a modular **agentic conversational system** using **ADK**, enabling structured multi-agent coordination, contextual memory routing, and adaptive tool invocation.
- Integrated the **Qwen** large language model into multi-agent reasoning and tool-augmented workflows, improving dialogue accuracy by **25%** through optimized orchestration and prompt design.

Edge Based Multimodal Smart Lock System | TensorFlow Lite, Raspberry Pi, OpenCV

- Constructed a **FaceNet-powered facial recognition module** on **Raspberry Pi**, achieving **92% accuracy** across 1,000+ real-world edge authentications.
- Integrated **gesture recognition** with OpenCV and multimodal fusion, improving lock reliability by **28%** under varying lighting and occlusion conditions.

Multimodal Voice & Sign Language Personal Assistant | TensorFlow, Porcupine, OpenCV

- Built a **voice and sign-language assistant** achieving **87% command accuracy** across 2,000+ test trials using **TensorFlow** and **OpenCV**.
- Automated **15+ system workflows** for scheduling, application launching, and OS control, improving task completion efficiency by **25%**.

SKILLS

Programming & Tools:	Python, SQL, Bash, Git, TypeScript, API Design, Agile/Scrum
ML & Deep Learning:	Transformers, Time-Series Modeling, PEFT/LoRA, Embedding Models, Neural Architecture Search (NAS)
Frameworks & Libraries:	PyTorch, TensorFlow, Keras, scikit-learn, pandas, NumPy, OpenCV, TFLite, ONNX, Flask/FastAPI
LLMs & Agentic AI:	LangGraph, LangChain, Google ADK, RAG Pipelines, Prompt Engineering, Multi Agent Systems
MLOps:	Docker, Kubernetes, MLflow, CI/CD, Model Serving, Experiment Tracking
Cloud & Data Infrastructure:	AWS (EC2, Lambda, S3), Google Cloud Platform (GKE, Cloud Run, Pub/Sub), Redis, Cassandra, PySpark